

Skill-3

INPUT-

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define INITIAL_SIZE 10

// Linked list node
struct Node {
    int data;
    struct Node *next;
};

// Demonstrate escape sequences
void escapeSequenceDemo() {
    printf("\n--- Escape Sequence Demo ---\n");
    printf("Hello\tWorld\n");
    printf("Line 1\nLine 2\n");
    printf("Quotation: \"Hello\"\n");
    printf("Backslash: \\n");
}

// Dynamic buffer input
char *readInput() {
    int size = INITIAL_SIZE;
    int length = 0;
    char *buffer = malloc(size * sizeof(char));

    if (buffer == NULL) {
        printf("Memory allocation failed!\n");
        exit(1);
    }

    int ch;

    while ((ch = getchar()) != '\n' && ch != EOF) {

        // Resize buffer before overflow
```

 ^G Help	 ^O Write Out	 ^F Where Is	 ^K Cut	 ^T Execute
 ^X Exit	 ^R Read File	 ^\ Replace	 ^U Paste	 ^J Justify

```
// Resize buffer before overflow
if (length >= size - 1) {
    size *= 2;

    char *temp = realloc(buffer, size * sizeof(char));

    if (temp == NULL) {
        free(buffer);
        printf("Memory reallocation failed!\n");
        exit(1);
    }

    buffer = temp;
}

buffer[length++] = ch;

buffer[length] = '\0';

return buffer;
}

// Linked list demonstration
void linkedListDemo() {
    struct Node *head = NULL;
    struct Node *second = NULL;
    struct Node *third = NULL;

    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    if (head == NULL || second == NULL || third == NULL) {
        printf("Memory allocation failed!\n");
        free(head);
    }
}
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```
bhagya_sri@DESKTOP-CDJB  ×  +  v
GNU nano 8.7.1 skill5 *
    printf("Memory allocation failed!\n");
    free(head);
    free(second);
    free(third);
    return;
}

head->data = 10;
head->next = second;

second->data = 20;
second->next = third;

third->data = 30;
third->next = NULL;

printf("\n--- Linked List ---\n");

struct Node *current = head;

while (current != NULL) {
    printf("%d -> ", current->data);
    current = current->next;
}

printf("NULL\n");

// Release memory correctly
free(head);
free(second);
free(third);
}

int main() {

    printf("=====\n");
    | printf("    INPUT BUFFER AND MEMORY DEMO\n");

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
printf("    INPUT BUFFER AND MEMORY DEMO\n");
printf("=====\\n");

escapeSequenceDemo();

printf("\\nEnter a command/text: ");

char *input = readInput();

printf("\\nYou entered: %s\\n", input);

free(input);

linkedListDemo();

printf("\\nProgram completed successfully.\\n");

return 0;
}
```

OUTPUT-

```

bhagya_sri@DESKTOP-CDJJB  x  +  v  -  □  ×
bhagya_sri@DESKTOP-CDJBPHA:~$ mkdir OS-SKILL-3
bhagya_sri@DESKTOP-CDJBPHA:~$ cd OS-SKILL-3
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ pw
pw: command not found
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ pwd
/home/bhagya_sri/OS-SKILL-3
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ nano skill3.c
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ gcc -Wall -Wextra -g skill3.c -o skill3
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ ./skill3
=====
      INPUT BUFFER AND MEMORY DEMO
=====

--- Escape Sequence Demo ---
Hello   World
Line 1
Line 2
Quotation: "Hello"
Backslash: \

Enter a command/text: hello world

You entered: hello world

--- Linked List ---
10 -> 20 -> 30 -> NULL

Program completed successfully.
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ ./skill.3
-bash: ./skill.3: No such file or directory
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ ./skill3
=====
      INPUT BUFFER AND MEMORY DEMO
=====

--- Escape Sequence Demo ---
Hello   World
Line 1
Line 2
Quotation: "Hello"
```

```
bhagya_sri@DESKTOP-CDJB  ×  +  ▾  -  □  ×  
Quotation: "Hello"  
Backslash: \  
  
Enter a command/text: This is a very long command that tests dynamic buffer  
resizing  
  
You entered: This is a very long command that tests dynamic buffer resizing  
  
--- Linked List ---  
10 -> 20 -> 30 -> NULL  
  
Program completed successfully.  
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ history  
1  sudo apt update  
2  sudo apt install build-essential  
3  gcc --version  
4  mkdir OSLab  
5  cd ~  
6  mkdir OSLab  
7  bhagya_sri@DESKTOP-CDJBPHA:~/OSLab$  
8  cd OSLab  
9  gedit hello.c  
10 nano hello.c  
11 gcc hello.c -o hello  
12 ./hello  
13 uname -a  
14 lscpu  
15 lsblk  
16 ps -ef  
17 top  
18 free -h  
19 lsblk  
20 #include <stdio.h>  
21 #include <unistd.h>  
22 int main()  
23 { printf("PID = %d\n", getpid());  
24 }  
25 Only one process exists.  
26 Introduce getpid().  
27 nano program1.c  
28 uname -a
```

```
bhagya_sri@DESKTOP-CDJB  ×  +  ▾  -  □  ×  
772  chmod 660 sample.txt  
773  ls -l sample.txt  
774  chmod 100 sample.txt  
775  ls -l sample.txt  
776  chmod 600 sample.txt  
777  ls -l sample.txt  
778  chmod 644 sample.txt  
779  ls -l sample.txt  
780  chmod 755 sample.txt  
781  ls -l sample.txt  
782  chmod 600 sample.txt  
783  ls -l sample.txt  
784  chmod 444 sample.txt  
785  ls -l sample.txt  
786  chmod 660 sample.txt  
787  ls -l sample.txt  
788  nano permission.c  
789  pwd  
790  mkdir practical-5  
791  cd practical-5  
792  pwd  
793  nano practical5.c  
794  pwd  
795  mkdir skill5  
796  cd skill5  
797  pwd  
798  nano skill5  
799  ls  
800  cat skill5.c  
801  gcc -Wall -Wextra -g practical5.c -o practical5  
802  gcc -Wall -Wextra skill5.c -o skill5  
803  mkdir OS-SKILL-3  
804  cd OS-SKILL-3  
805  pw  
806  pwd  
807  nano skill3.c  
808  gcc -Wall -Wextra -g skill3.c -o skill3  
809  ./skill3  
810  ./skill.3  
811  ./skill3  
812  history
```

812 history

bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3\$ valgrind --leak-check=full ./skill3

Command 'valgrind' not found, but can be installed with:

sudo snap install valgrind # version 3.26.0, or

sudo apt install valgrind # version 1:3.26.0-0ubuntu1

See 'snap info valgrind' for additional versions.

bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3\$ sudo apt update

[sudo: authenticate] Password:

Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease

Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]

Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]

Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [424 kB]

Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]

Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [549 kB]

Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.9 kB]

Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]

Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [128 kB]

Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]

Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]

Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]

Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [252 kB]

Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]

Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [84.4 kB]

Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [361 kB]

Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]

Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [371 kB]

Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted Transl


```
Get:23 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Co
mponents [1056 B]
Fetched 3417 kB in 7s (522 kB/s)
41 packages can be upgraded. Run 'apt list --upgradable' to see them.
bhagya_sri@DESKTOP-CDJBPHA:~/OS-SKILL-3$ sudo apt install valgrind
Installing:
  valgrind

Installing dependencies:
  gdb                libdebuginfod-common  libipt2
  libbabeltrace1     libdebuginfod1t64         libsource-highlight-common
  libc6-dbg          libdw1t64                 libsource-highlight4t64

Suggested packages:
  gdb-doc  gdbserver  valgrind-mpi  kcachegrind

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 41
  Download size: 27.1 MB
  Space needed: 111 MB / 1023 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-com
mon all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0
.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 am
d64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64
amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1
.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlig
ht-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlig
ht4t64 amd64 3.1.9-4.3build2 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2u
buntu1 [4173 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg
amd64 2.43-2ubuntu2.3 [6407 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1
```

```
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
bhagya_sri@DESKTOP-CDJBPH:~/OS-SKILL-3$ valgrind --version
valgrind-3.26.0
bhagya_sri@DESKTOP-CDJBPH:~/OS-SKILL-3$ valgrind --leak-check=full ./skill3
==1375== Memcheck, a memory error detector
==1375== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==1375== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==1375== Command: ./skill3
==1375==
=====
      INPUT BUFFER AND MEMORY DEMO
=====

--- Escape Sequence Demo ---
Hello World
Line 1
Line 2
Quotation: "Hello"
Backslash: \

Enter a command/text: hello world

You entered: hello world

--- Linked List ---
10 -> 20 -> 30 -> NULL

Program completed successfully.
==1375==
==1375== HEAP SUMMARY:
==1375==    in use at exit: 0 bytes in 0 blocks
==1375== total heap usage: 7 allocs, 7 frees, 2,126 bytes allocated
==1375==
==1375== All heap blocks were freed -- no leaks are possible
==1375==
==1375== For lists of detected and suppressed errors, rerun with: -s
==1375== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bhagya_sri@DESKTOP-CDJBPH:~/OS-SKILL-3$ |
```